Jupiter's hidden depths: Simulation suggests planet holds 1.5 times more oxygen than the sun

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Des tempêtes gigantesques tourbillonnent à travers la surface de Jupiter. Ces tempêtes ont rendu impossible de voir ce qui se trouve en dessous, mais une nouvelle simulation menée par un scientifique de l'UChicago ajoute une nouvelle profondeur à notre compréhension. Crédit: NASA/JPL-Caltech/SwRI/ MSSS, Traitement d'image par Kevin M. Gill, CC PAR

Des nuages spectaculaires tourbillonnent à travers la surface de Jupiter. Ces nuages contiennent de l'eau, tout comme celle de la Terre, mais sont beaucoup plus denses sur la géante gazeuse – si épaisse qu'aucun vaisseau spatial n'a été capable de mesurer exactement ce qui se trouve en dessous.

Mais une nouvelle étude menée par les scientifiques de l'Université de Chicago et du Jet Propulsion Lab nous a donné un regard plus approfondi sur la planète en créant le modèle le plus complet à ce jour de l'atmosphère de Jupiter.

Entre autres choses, l'analyse aborde une question de longue date sur la quantité d'oxygène que contient la géante gazeuse: elle estime que Jupiter a environ une fois et demie plus d'oxygène que le soleil. Cela aide les scientifiques à réduire l'image de la façon dont toutes les planètes du système solaire se sont formées.

"Il s'agit d'un débat de longue date dans les études planétaires", a déclaré Jeehyun Yang, chercheur postdoctoral à l'UChicago et premier auteur sur le papier. "C'est un témoignage de la façon dont la dernière génération de modèles informatiques peut transformer notre compréhension d'autres planètes."

The study was published Jan. 8 in *The Planetary Science Journal*.

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A compilation of the Hubble Space Telescope's snapshots of the giant planet Jupiter as it moves. The planet's real rotation rate is nearly 10 hours, and Jupiter's stripes are actually bands of clouds and storms. Credit: NASA, ESA, Amy Simon (NASA-GSFC)

Clouds and chemistry

We've known about Jupiter's stormy skies for at least 360 years—that's when astronomers using early telescopes documented a curious, large permanent blotch on Jupiter's surface.

The Great Red Spot is a gigantic storm, twice the size of Earth, that has swirled for centuries. It's just one of many on the planet, as fierce winds and deep clouds mean the entirety of Jupiter's surface is covered in a kaleidoscope of storms.

What we don't know is precisely what lies beneath those storms. The clouds are so thick that NASA's Galileo spacecraft lost contact with Earth as it plunged into the deeper atmosphere in 2003. The next mission to visit Jupiter, Juno, is currently cataloging the planet from a safe distance in orbit.

These measurements from orbit can tell us the components in the upper atmosphere: ammonia, methane, ammonium hydrosulfide, water and carbon monoxide, among others. Scientists have combined this with knowledge about chemical reactions to build models of Jupiter's deep atmosphere.

But studies have disagreed about certain points, such as how much water—and thus oxygen—the planet contains. Yang saw an opportunity to apply a new generation of chemical modeling to the knotty question.

The chemistry of Jupiter's atmosphere is incredibly complex. Molecules travel between the extremely hot conditions deep in the atmosphere and the cooler upper regions, changing phases and rearranging into different molecules, via thousands of different types of reactions. But the behavior of clouds and droplets has to be accounted for, too.

To better capture all these phenomena, Yang worked with a team of scientists to incorporate both chemistry and hydrodynamics into one model—a first.

"You need both," Yang said. "Chemistry is important but doesn't include water droplets or cloud behavior. Hydrodynamics alone simplifies chemistry too much. So, it's important to bring them together."

Elemental questions

Among the findings is a new calculation for how much oxygen Jupiter has. According to their analysis, Jupiter likely has about one and a half times more than the sun.

For decades, scientists have been arguing about this number. A major recent study had put it much lower, at only a third of the sun's.

But knowing this statistic is particularly relevant for understanding how our solar system formed.

All of the elements that make up planets—and us—are the same stuff that makes up the sun. But there can be differences in the amounts of these materials, and we can use those clues to piece together how the planets must have formed.

For example, did Jupiter form in the same place where it is now, or did it form closer or further away and drift over time? Clues can come from the fact that much of the oxygen in the planet is bound up in water, which will freeze—and behave differently—if it's too far away from the warmth of the sun. Ice is easier for planets to accumulate than water vapor.

In turn, knowing more about which conditions create which kinds of planets can help us as we search for habitable planets beyond our own solar system.

The model also suggested that Jupiter's atmosphere likely circulates up and down much more slowly than long believed.

"Our model suggests the diffusion would have to be 35 to 40 times slower compared to what the standard assumption has been," said Yang. For example, it would take a single molecule several weeks to move through one layer of the atmosphere, rather than hours.

"It really shows how much we still have to learn about planets, even in our own solar system," Yang said.

Publication details

Jeehyun Yang et al, Coupled 1D Chemical Kinetic Transport and 2D Hydrodynamic Modeling Supports a Modest 1–1.5× Supersolar Oxygen Abundance in Jupiter's Atmosphere, *The Planetary Science Journal* (2026). DOI: 10.3847/psj/ae28d5

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